# Lab 3: Component Modelling & Architectural Pattern Selection

## Self-Service Coffee Kiosk System — Architecture Justification

**Architecture Selection:** We chose a layered architecture for the Self-Service Coffee Kiosk System.

**Reasons for the choice:** First, the kiosk has a focused workflow—touchscreen selection, order validation, credit-card payment, and receipt printing. Separating presentation, order logic, and service/data access keeps responsibilities clear and makes the system easier to test and maintain. Second, the kiosk must integrate with the touch screen and receipt printer while keeping menu items and prices available to the order process. A dedicated printer adapter and menu/pricing store isolate these dependencies so hardware or storage changes do not require rewriting the whole application.

**Security advantage:** The Payment Service isolates card-processing logic from the touchscreen and order UI. Payment requests should pass through this component, full card details should not be stored by the kiosk, and an order should be marked paid only after authorization is confirmed.

**Performance benefit:** Menu and price data can be loaded locally and queried without a network round trip for every selection. Keeping the order workflow within the kiosk process reduces inter-component network latency, while the printer adapter handles printing separately from UI interaction.